

don't fully realize the returns on high-performing assets, we also avoid the disappointment and volatility of negative extremes.

So regardless of whether you are looking at three or three hundred stocks or funds, there's an implicit trade-off when you diversify. Still, there is universal agreement that diversification is an essential investment strategy. To gain a deeper understanding of why and how the statistical benefits of diversification outweigh the potential gains of investing in a single outstanding stock, you have to consider the risk dimension.

Why Do Prices Move Up and Down So Much?

Generally speaking, when we are considering a given investment asset, the industry assesses the riskiness or volatility of this asset's returns by using something the statisticians call the standard deviation of the asset's returns. This statistic is a measure of how widely the investment returns differ from the average. Figure 3.2 illustrates what this means graphically based on 78 years of return values from the SP500 total return index and long-term bonds. You can see that in the case of the SP500 total return, each return bin or category along the horizontal x-axis contains recorded returns, even if with a low frequency. The returns are widely dispersed, showing a greater probability of achieving very high gains and devastatingly high losses. This is the characteristic shared by all high-risk or volatile investments. Conversely, long-term bond returns over the same period are clustered around the expected or average return for the asset class at the middle of the horizontal axis, with few if any observations in the outer return categories. This low volatility or dispersion is typical of low-risk or low-volatility investments.

Now even more specifically, one must recognize that the risk associated with a stock is really a combination of two types of very distinct risks. One is called general market risk; the other is called individual security risk. Let me briefly explain what I mean by these types of risk, and why it's important to keep track of both.

General market risk, or systemic risk, refers to risk associated with changes to stock prices that have nothing to do with an individual company, but are instead due to changes in the overall economy. For example, if the U.S. Federal Reserve decides to raise interest rates